

PROJECT HELM

Helm Phase 0: Solution Architecture & MVP Delivery Plan

Client meeting, 9 October 2026

Cyclone

Draft for discussion · v0.1 · evidence verified 8 October 2026

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How to read this document

Part I (sections 1–8) is for Business and the client. It explains what Members can do, which platform supplies each capability, what Cyclone builds, how fast an MVP can credibly launch and which decisions are needed at the meeting. It does not require the technical sections.

Part II (sections 9–15) is for the Tech teams. It refines the PRD and specifies the architecture, data flows, genealogy model, backlog, launch gate and responsibilities.

The appendix holds evidence links, assumptions, unresolved details and questions for providers.

Labels used throughout:

Label	Meaning
Verified	Read in the provider's own documentation, API specification, contract source or pricing page on 8 October 2026 (links in Appendix A).
Correspondence	Stated in provider correspondence supplied by Business; not independently verified.
Recommended	Cyclone's proposed design or default. Not yet approved.
TBD	Assumption or open item. Needs a decision or provider confirmation.

The stack is confirmed: **PillarsHub** (MLM core), **Privy** (login and wallets), **Enzyme Onyx** (vault infrastructure), **Chatwoot** (support), Cyclone's **Tap Prediction game and points**, and a custom **Helm Web App** with its backend. This document does not compare or reselect providers. Terrel's PRD was not available; requirements that could not be checked against it are listed in Appendix B.

PART I: BUSINESS AND CLIENT

1 Executive overview

The confirmed stack can deliver Phase 0. Every capability the draft PRD asks for is available from a confirmed provider or needs only configuration and a small amount of Cyclone integration work. No evidenced blocker requires changing a provider.

What Members will be able to do in the MVP:

1. Join Helm through an invite link, with their Sponsor recorded permanently from day one.
2. Log in with email or social login and receive a Helm wallet automatically. Only the Member can move its funds.
3. Fund that wallet by sending USDC on one network from any exchange or wallet.
4. Read the vault terms, deposit into the approved vault, see their confirmed position, and request a Redemption.
5. Play the existing Tap Prediction game and earn Points, which have no cash value.
6. Chat with support from inside the app, with support seeing who they are but never their keys.

Why this is the fastest credible path:

- **Cyclone builds only the glue:** the member experience, one backend that connects the providers, a ledger of member identities, referrals and events, and the reconciliation and admin views that make money flows auditable.
- **Each provider does what it already does:** Privy runs login and wallets; PillarsHub stores the genealogy and later calculates Commissions; Onyx issues vault shares and runs deposit and redemption queues; Chatwoot handles conversations.
- **Direct crypto funding** avoids any payment-provider onboarding on the critical path.

What can start immediately (no provider or legal answer needed): the Helm Web App shell, Privy login and wallet creation, the member and referral data model, the PillarsHub staging sync, the Chatwoot widget, chain indexing, and the full deposit and redemption journey on Enzyme's Sepolia test network.

What gates live funds (none of these blocks the build):

1. **The vault strategy and its Manager.** Onyx is vault infrastructure, not a strategy. Someone must be appointed to run the strategy, report the vault's value and return liquidity for Redemptions. Verified.
2. **Vault ownership and Enzyme's upgrade rights.** The vault Owner and Admins are fully trusted by the protocol, and Enzyme can upgrade vault contracts. The owning entity, a multisig and the upgrade terms in the licence agreement must be settled. Verified.
3. **Enzyme's licence agreement (MLA) and commercial terms.** Production use requires a commercial agreement. The public price differs from the quoted terms. Verified.
4. **Legal applicability review** of the operating entity, launch countries, KYC tier and vault terms. No confirmed provider supplies usable KYC or screening for this model.
5. **Provider acceptance of the business model**, in writing, from Privy, Enzyme and PillarsHub.

Earliest credible schedule (estimate, not a commitment): a working end-to-end test slice about 3 weeks after kickoff, and an MVP ready for a controlled launch in about **8–12 weeks**, provided the vault, licence and legal answers arrive by around week 6. Section 6 gives the assumptions.

2 Confirmed stack: what each system provides

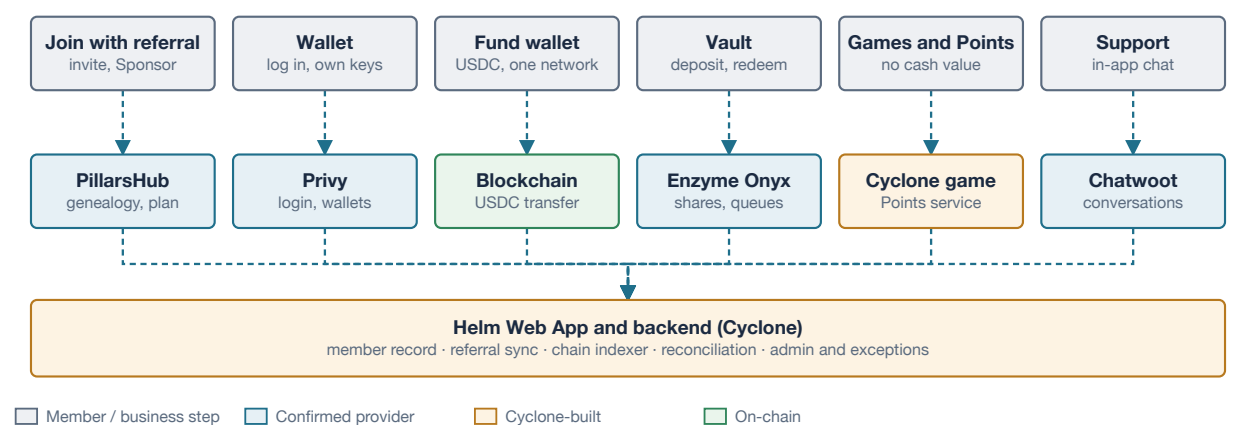


Figure 1: Business capabilities, the platform that provides each, and what Cyclone builds.

Capability	Provided by	Cyclone builds	Status
Join and log in; one stable Helm identity	Privy (login, identity token)	Member record, Privy-to-Helm mapping	Feasible
Wallet	Privy embedded wallet, owned by the Member; gas sponsored by Helm	Wallet screens; balance display	Feasible
Referral and genealogy	PillarsHub (customer with sponsor, Unilevel tree, full-tree export)	Referral capture, validation, sync, reconciliation	Feasible with configuration
Wallet funding	Blockchain (USDC transfer to the Member's address)	Address display, confirmation tracking, screening	Feasible
Vault deposit and redemption	Enzyme Onyx (allowlisted deposit queue, redemption queue, shares, public read API)	Terms screens, transaction flows, status tracking, allowlist updates	Feasible with configuration; live use blocked pending strategy, MLA and ownership
Games and Points	Cyclone Tap Prediction and points service	Account linking, verified events, Points display	Feasible pending game API documentation
Support	Chatwoot Cloud (web widget with identity validation)	Widget embedding, identity signing	Feasible
Operations	Provider dashboards (Onyx Admin App, PillarsHub Portal, Chatwoot)	Helm admin: exceptions, reconciliation, audit log	Feasible

3 The Member journey

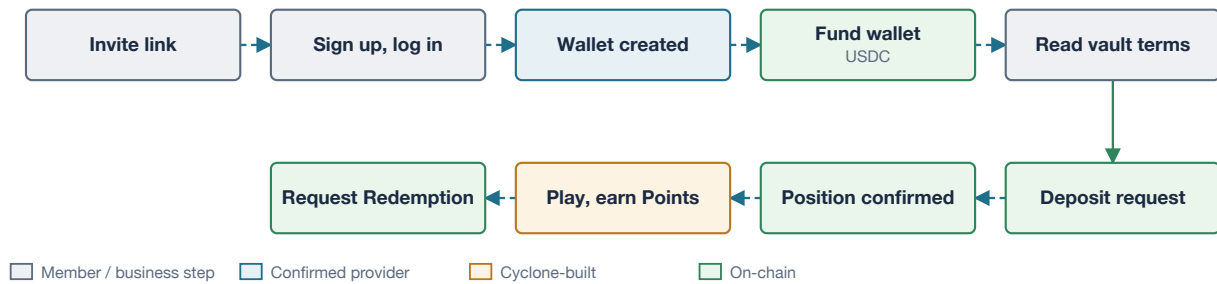


Figure 2: Main Member journey. Wallet funding and vault deposit are separate steps; Points are separate from money.

1. **Invite and signup.** The Member opens a Sponsor's invite link and logs in with Privy. Helm records a pending referral and confirms it once PillarsHub accepts it.
2. **Wallet.** Privy creates an embedded wallet. Only the Member can sign with it; Helm and Privy cannot move its funds.
3. **Wallet funding.** Helm shows one supported network and asset (recommended: USDC on Arbitrum) and the Member's address. Funding moves money into the Member's own wallet; it is **not** an investment and creates no Commission eligibility.
4. **Vault deposit.** The Member reads the vault terms, fees and risks, then signs two transactions: permission for the vault to take the amount, and the deposit request. The request is pending until vault operations process it at the next valuation. Only then does the Member hold vault shares.
5. **Position.** Helm shows the confirmed share balance and its value as of the vault's last valuation, with the valuation date.
6. **Engagement.** The Member plays Tap Prediction and earns Points. Points have no cash value and do not affect vault positions or Commissions.
7. **Redemption.** The Member requests a Redemption. It is processed by vault operations when liquidity is available; Helm shows each stage. There is no protocol-guaranteed processing time, so Helm must publish an operational target.

4 Phase 0 scope

In scope	Deferred unless Business requires it at launch
One network and one funding asset (recommended: USDC on Arbitrum One)	Integrated purchase or top-up (Paymenture or similar)
One approved Onyx vault, asynchronous deposits, queued Redemptions	Additional chains, assets or vaults; instant deposits
Privy embedded wallets with sponsored gas	Server wallets, automated strategies, policies
Referral capture and full genealogy in PillarsHub	Commission accrual and payouts (decision D2)
Existing Tap Prediction game and Points	New games, tokens, Points redemption for value
Chatwoot web chat	Telegram and WhatsApp support
Helm admin for exceptions, reconciliation and audit	Duplicating provider dashboards
Sanctions screening of wallet addresses; country eligibility; Member terms	Full KYC unless counsel requires it (decision D6)

5 Feasibility summary

Area	Verdict	Key evidence	What remains
Login and identity	Feasible	Privy issues verifiable identity tokens; one Privy user maps to one Helm member. Verified.	Confirm Privy accepts the business model.
Wallet	Feasible	User-owned embedded EVM wallet; Arbitrum supported; native gas sponsorship; key export. Verified.	Losing the only login method loses the wallet; offer a second login method.
Genealogy	Feasible with configuration	Customer records take a Helm-defined ID and the sponsor at creation; the whole tree can be listed with an as-of date. Verified.	Confirm auto-placement in the tree, depth limits, sponsor-correction method, token permissions.
Commissions (Phase 1 readiness)	Feasible	Payable depth is set in the plan, not the tree; custom volume types exist; payouts are a separate step. Verified.	Decide whether anything accrues in Phase 0.
Wallet funding	Feasible	Standard USDC transfer; Helm indexes the chain.	Choose network and asset.
Vault deposit and redemption	Feasible with configuration; live use blocked	Allowlisted deposit queue; Member must sign; admin executes after valuation; queued Redemptions; read-only API. Verified.	Strategy and Manager, MLA, ownership, upgrade terms, operating process.
Games and Points	Feasible pending evidence	No game documentation supplied.	API documentation, event verification and abuse controls.
Support	Feasible	Web widget with signed identity; Cloud plans from \$19 per agent per month. Verified.	Choose plan; confirm US data hosting is acceptable.
Compliance tooling	Gap within the stack	No confirmed provider offers usable KYC or screening for this model.	Counsel decision; add a screening service at the Helm backend if required.

6 MVP delivery plan

Delivery is organised so that engineering never waits for provider or legal answers. Those answers gate **live funds**, not the build.

Stage	Duration (estimate)	Outcome	Gate to next stage
0. Access and decisions	Week 1	Meeting decisions recorded; PillarsHub staging, Privy app, Chatwoot, Onyx Sepolia vault and game docs in hand	Access confirmed
1. Thin end-to-end slice	Weeks 1–3	On test networks: invite → signup → wallet → referral accepted in PillarsHub → funding → deposit request → executed position → Redemption; one game-to-Points flow	Slice demonstrated
2. MVP build	Weeks 3–8	Full member UI, admin and exception handling, reconciliation, screening, terms, support, monitoring	Feature-complete on test networks
3. Production readiness	Weeks 6–10 (overlaps)	Production vault deployed and handed over, security review of Helm's deployment, legal sign-off, operations rehearsal	Launch gate (section 14) passed
4. Controlled launch	From about week 8–12	Invitation-only cohort with exposure limits	Agreed checks pass, then expand

Assumptions behind these ranges: a Cyclone team of about five engineers plus QA and a delivery lead; PillarsHub, Privy and Chatwoot access in week 1; Enzyme's test vault available in week 1; game API documentation available by week 2; vault strategy, MLA, ownership decisions and legal answers by around week 6. PillarsHub only launches clients Tuesday to Thursday, 9:00–16:00 US Mountain Time, which constrains the go-live date. Verified. No date in this table is a commitment.

Indicative running costs (USD per month unless stated; prices verified on 8 October 2026 where marked):

Item	Cost	Status
Privy	Free up to 499 monthly active users; \$299 up to 2,499; \$499 up to 9,999. The free tier includes \$1M of transaction volume per month.	Verified; overage and paid-tier volume limits TBD
Gas sponsorship	Network gas plus a Privy fee	TBD (usage-based)
Enzyme Onyx	Public page: \$2,000 per month, or 20% of vault fees with a \$6,000 annual minimum. Correspondence: \$5,000 deployment plus 0.25% of AUM, or 20% revenue share.	Verified (public) vs correspondence; reconfirm in the MLA
PillarsHub	Not published	TBD
Chatwoot Cloud	\$19 or \$39 per agent (Startups, Business); \$99 for Enterprise with audit logs and SSO	Verified
Sanctions address screening	Free Chainalysis sanctions API at pilot scale	Verified (discovery research)
KYC service, if counsel requires it	About \$0.33–\$1.85 per check at published prices	Verified (discovery research)
Security review of Helm's deployment	Quotation needed	TBD

7 Security and compliance summary

- **Custody.** Member wallets are self-custodial: only the Member signs. Neither Cyclone nor Privy can move Member funds. Verified (Privy). The vault is different: its Owner and Admins can set the share value and withdraw vault assets to the strategy wallet, and Enzyme can upgrade the contracts. Who holds those roles, under what controls, is the main custody question for Business and counsel.
- **Admission to the vault can be controlled on-chain.** Onyx lets only allowlisted wallet addresses deposit. Helm adds a Member's address only after its eligibility checks pass. Verified.
- **No provider in the stack supplies usable KYC or sanctions screening for this model.** Privy's KYC runs on Bridge, whose terms prohibit MLM. If counsel requires identity checks, a screening service is added at the Helm backend. This adds a service; it does not change the stack.
- **Minimum before real funds, whatever counsel concludes:** screening of every Member wallet, funding source and Redemption destination address against sanctions lists; country eligibility; Member terms, risk and fee disclosures and a privacy notice; a written procedure for flagged users and transactions.
- **Points and referrals.** Points have no cash value and are kept separate from money and Commissions. The referral model and any link between Points and deposits still need counsel's review; non-cash status alone is not an exemption.
- **Provider audits do not cover Helm's deployment.** Helm's vault configuration, keys, valuation process and integrations need their own security review before funds.

8 Decisions needed at the meeting

#	Decision	Recommended answer	Why it matters
D1	Network and funding asset	Arbitrum One, native USDC; test on Ethereum Sepolia	Fixes wallet, vault and indexing work
D2	Commissions in Phase 0	Record the genealogy and vault events, but no Commission accrual or payout	Avoids an implied payout obligation; Phase 1 can backfill from Helm's records
D3	Vault strategy, Manager and ownership	Business appoints the Manager; vault Owner is a multisig controlled by Business	Gates live funds
D4	Redemption service level	Publish an operational target (for example, within 5 business days, subject to liquidity)	Members see stages, not a guarantee
D5	Integrated top-up at launch	No; direct crypto funding	Removes a provider from the critical path
D6	Identity checks	Counsel decides the tier; build the gates now; screen addresses from day one	Avoids rework if KYC is required
D7	Points rules	No cash value, not transferable, not awarded for deposit amounts	Keeps Points separate from investment activity
D8	Members without a valid Sponsor	Attach to a company root account, flagged for review	Keeps the genealogy complete

PART II: TECHNICAL IMPLEMENTATION

9 Refined high-level PRD

9.1 Objectives

1. Onboard Members with one stable identity and a usable, self-custodial wallet.
2. Enable direct crypto Wallet Funding and participation in one approved vault, including a practical Redemption journey.
3. Reuse the existing Tap Prediction game and Points to drive engagement.
4. Capture the complete, portable genealogy from the first signup, so Phase 1 Commissions need no identity rebuild.

9.2 Personas

- **Member:** joins through a referral, funds a wallet, deposits, plays, redeems, asks for support.
- **Operations administrator:** reviews referral and funding exceptions, maintains the vault allowlist, monitors queues and reconciliation, handles escalations. Uses Helm admin plus provider dashboards.
- **Vault operator** (Business-appointed): updates the vault valuation, executes deposit and redemption queues, manages liquidity. Uses the Onyx Admin App.
- **Support agent:** answers Members in Chatwoot; cannot move funds, change Sponsors or award Points.
- **Business reviewer:** reads funnel, participation, referral and engagement reports.

9.3 Functional requirements and acceptance criteria

ID	Requirement	Acceptance criteria
FR-1	Authenticate with Privy and resolve one Helm member	Repeat logins with any linked method resolve to the same Helm member ID; the backend verifies every Privy token; duplicate-person cases go to an admin queue
FR-2	Create one embedded EVM wallet per Member	Wallet exists after first login; address is read from Privy server-side; the Member can sign a test transaction; recovery and export behaviour is explained in the app
FR-3	Capture a Referral Request at signup	Invite code or link is captured before login and bound to the new member; self-referral is rejected; missing or invalid Sponsor follows rule D8
FR-4	Accept the Sponsor Edge in PillarsHub	Customer created with the Helm-defined ID and Sponsor; node read back with the expected upline; edge marked accepted only after read-back; retries never create duplicates
FR-5	Show supported funding network and asset	Only one network and asset are offered; wrong-network warning is shown; funding moves through submitted → confirmed (N blocks) → credited; failed or unsupported transfers go to exceptions
FR-6	Screen addresses	Member wallet on creation, each funding source and each Redemption destination are screened; a hit blocks the vault allowlisting or the action and opens a case
FR-7	Gate vault access	Member must accept current vault terms (versioned) and pass eligibility checks; only then is the wallet added to the Onyx allowlist
FR-8	Vault deposit	Member signs exact-amount approval and deposit request; Helm tracks approval → request → pending → executed or cancelled; position counts only after execution
FR-9	Position display	Shares and value as of the vault's last valuation, with the valuation timestamp; matches chain state
FR-10	Redemption	Member signs a redemption request; Helm shows pending, awaiting liquidity, executed, cancelled; asset arrives in the Member's wallet; position and business events adjust
FR-11	Games and Points	Linked game account; verified events award Points once; rules and daily limits visible; Points shown separately from money
FR-12	Support	Verified chat identity; agents see Helm member ID and non-sensitive status only; logout resets the chat

ID	Requirement	Acceptance criteria
FR-13	Admin and reconciliation	Daily reconciliation of genealogy, funding, vault positions and Points; mismatches appear in an exception queue with an audit trail
FR-14	Reporting	Signups, funded Members, confirmed vault participation, referral completeness and engagement, with freshness shown and no double counting of funding and deposits

Three-level Commissions: not required for launch unless Terrel's PRD says otherwise (decision D2). The genealogy is captured at full depth regardless.

9.4 Operational requirements

- Every integration write is idempotent from Helm's side, retried with backoff and reconciled daily.
- No private key, seed phrase or privileged provider credential reaches the browser, logs or support payloads.
- Separate test and production environments for every provider, with separate credentials.
- Monitoring and alerts for: stale vault valuation, deposit or redemption requests pending beyond target, reconciliation breaks, sync failures, screening hits.

9.5 Success metrics (targets TBD)

Time to first approved live deposit; signup completion; signup-to-funded and funded-to-deposited conversion; referral attribution completeness (target 100% of accepted signups); reconciliation exceptions per week; game participation and repeat engagement; support first-response and resolution time.

10 Solution architecture

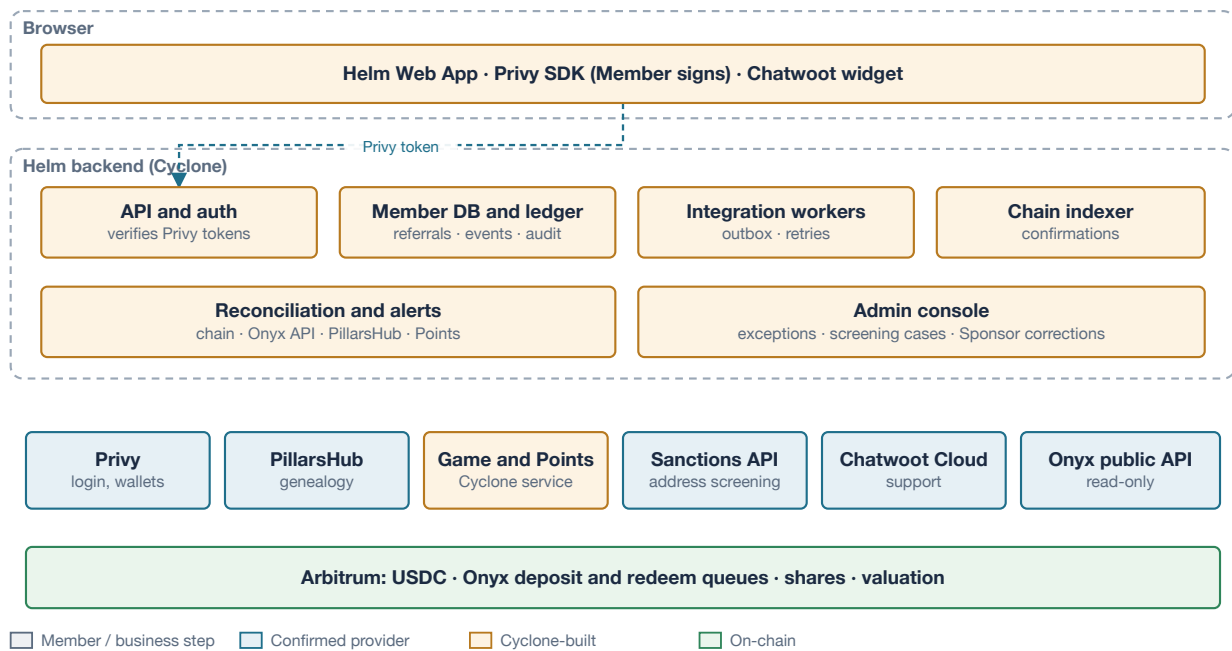


Figure 3: Technical architecture and trust boundaries.

10.1 Components and why each is needed

Component	Responsibility	Why custom
Helm Web App	Member and admin UI; Privy login; signing via the Member's embedded wallet; Chatwoot widget	One Member experience across providers
API and auth	Verifies Privy access tokens; resolves Helm member ID; serves the app	Providers cannot share one identity on their own
Member DB and ledger	Members, provider-ID mappings, wallets, Referral Requests, Sponsor Edges and corrections, business events, consent records, audit log	Single place to reconcile identities and money events
Integration workers	Outbox jobs to PillarsHub, game service, screening and allowlist; retries and dead letters	No provider offers idempotency keys across all calls
Chain indexer	Indexes USDC transfers to Member wallets and Onyx queue, shares and valuation events at confirmation depth	Privy production webhooks need Enterprise; Onyx has no webhooks. Verified
Reconciliation	Daily checks against chain, Onyx API, PillarsHub tree and Points	Provider data can drift; must be provable
Admin console	Exceptions, screening cases, allowlist status, Sponsor corrections with approval	Minimal; provider dashboards cover the rest

10.2 Data authority

Object	Authoritative record	Helm keeps
Helm member identity and provider-ID mapping	Helm DB	Master
Login identity	Privy user (DID)	Mapping only
Wallet address	Privy (server-side read) and chain	Association with first-seen time
Accepted genealogy	PillarsHub tree (customer enroller plus node upline)	Pending requests, accepted-edge copy, audit log of corrections
Wallet funding	Blockchain	Indexed, confirmed events

Object	Authoritative record	Helm keeps
Vault deposit requests, positions, Redemptions	Blockchain (Onyx contracts)	Indexed events; Onyx API as cross-check
Vault valuation	Onyx valuation contract (set by vault operator)	Displayed with timestamp
Points balance	Cyclone points service	Display copy
Conversations	Chatwoot	Link to member ID
Terms acceptance, screening results, eligibility	Helm DB	Master

10.3 Credentials, signing and custody boundaries

Actor	Can sign or act	Cannot	Notes
Member	Wallet transfers, vault approval, deposit and redemption requests, cancellations	Admin actions	Privy embedded wallet; Onyx requires the Member's own address to sign its requests. Verified
Helm backend	PillarsHub writes (customers, nodes, sources) with a limited token; game API; screening API; vault allowlist updates through a list-owner key	Move Member funds; vault admin actions	List-owner key held in a key management service; owns only the allowlist contract (to confirm with Enzyme)
Vault Owner (Business)	Add and remove Admins; any admin action	—	Recommended: Safe multisig with Business signers
Vault Admin / operator (Business)	Update valuation; execute queues; move assets to the strategy wallet; set fees	—	Fully trusted by the protocol. Verified
Strategy Manager (Business-appointed)	Runs the strategy outside Onyx; returns liquidity	—	Custody of the strategy wallet sits outside Onyx. Verified
Enzyme	Deploys; upgrades contracts (global owner)	—	Upgrade governance must be in the MLA. Verified
Privy	Runs the signing enclave	Sign for the Member	Privy "is never an authorized signer". Verified
Support agent	Reply in Chatwoot; read-only member status	Funds, Sponsors, Points	Enforced in Helm admin

PillarsHub tokens are environment-bound and do not expire; they can be made read-only. Recommended: a read-only token for reconciliation and a separate write token, with destructive areas denied if PillarsHub allows it (TBD). Payout, period and bonus-release endpoints must never be called by Helm in Phase 0.

11 End-to-end data flows

Every flow distinguishes **submitted** (Helm has a request or transaction hash), **confirmed** (provider acknowledged or block confirmations reached) and **synchronised** (Helm's copy matches the authority after reconciliation). Webhooks are treated as hints; polling and indexing are authoritative.

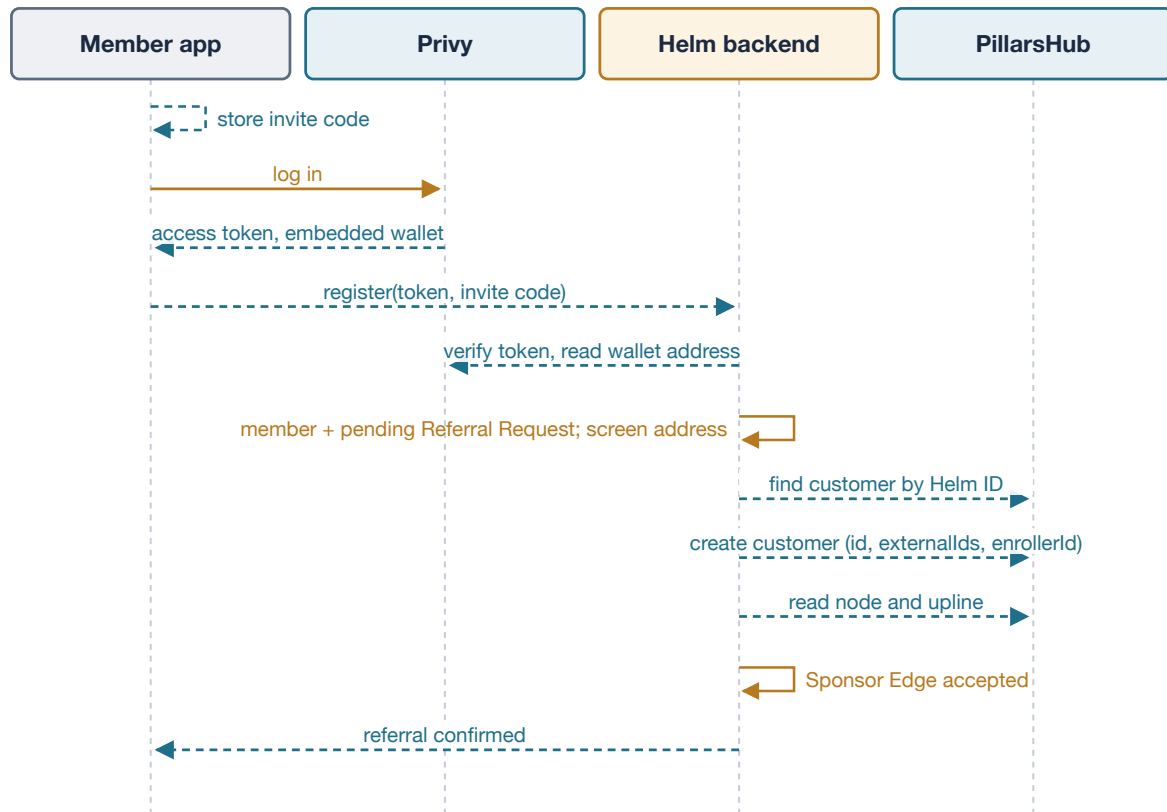


Figure 4: Invite, signup, wallet and accepted genealogy.

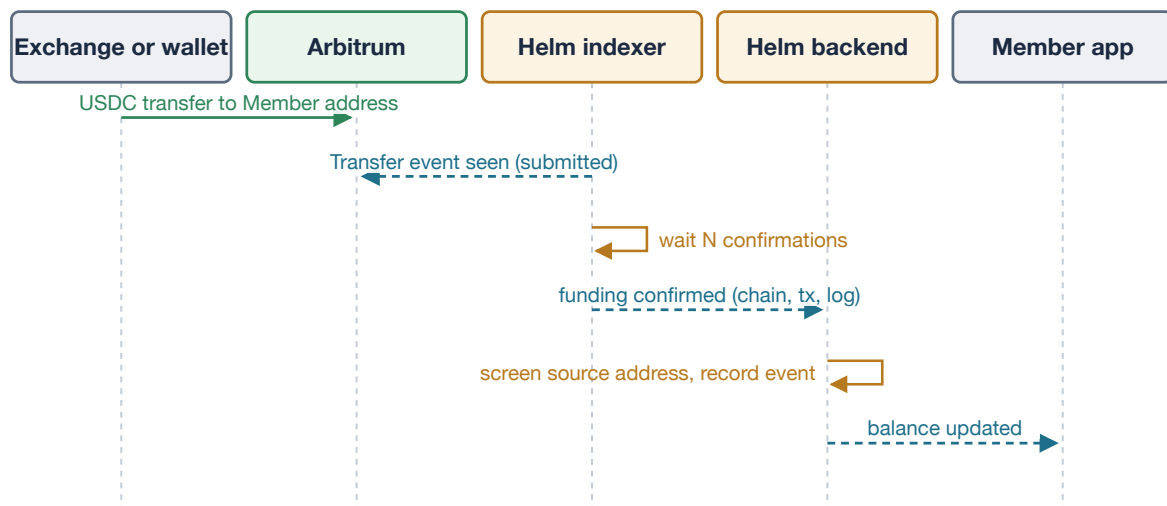


Figure 5: Direct wallet funding.

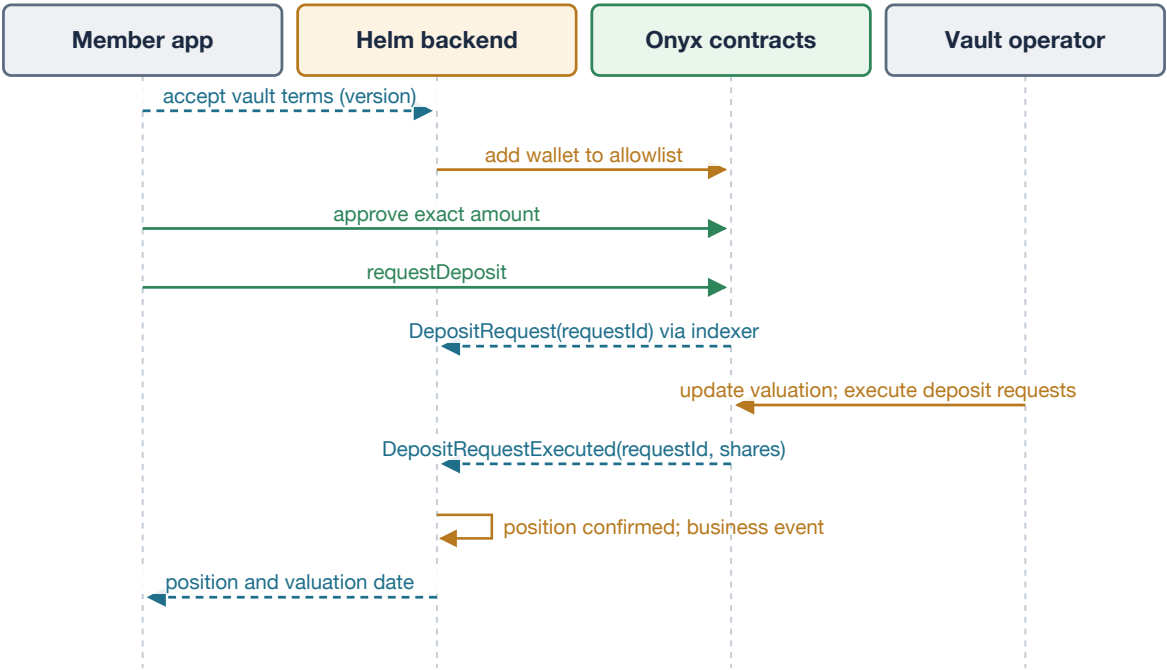


Figure 6: Vault deposit, execution and position.

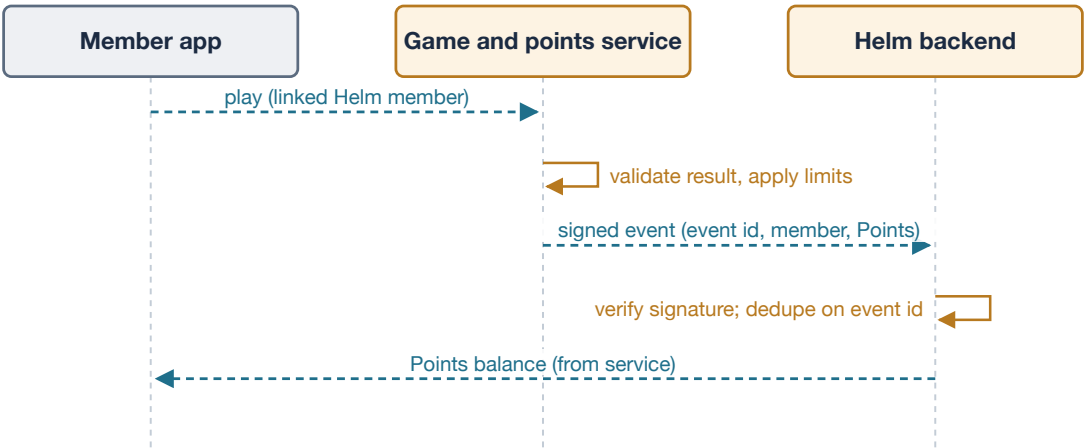


Figure 7: Game activity to Points.

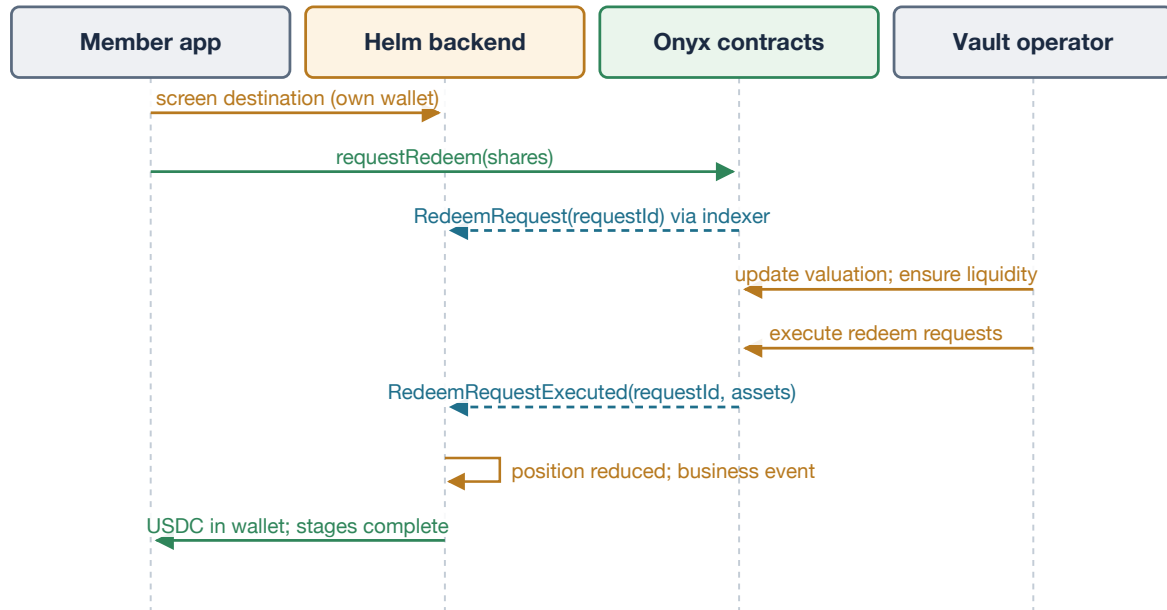


Figure 8: Redemption request to settlement.

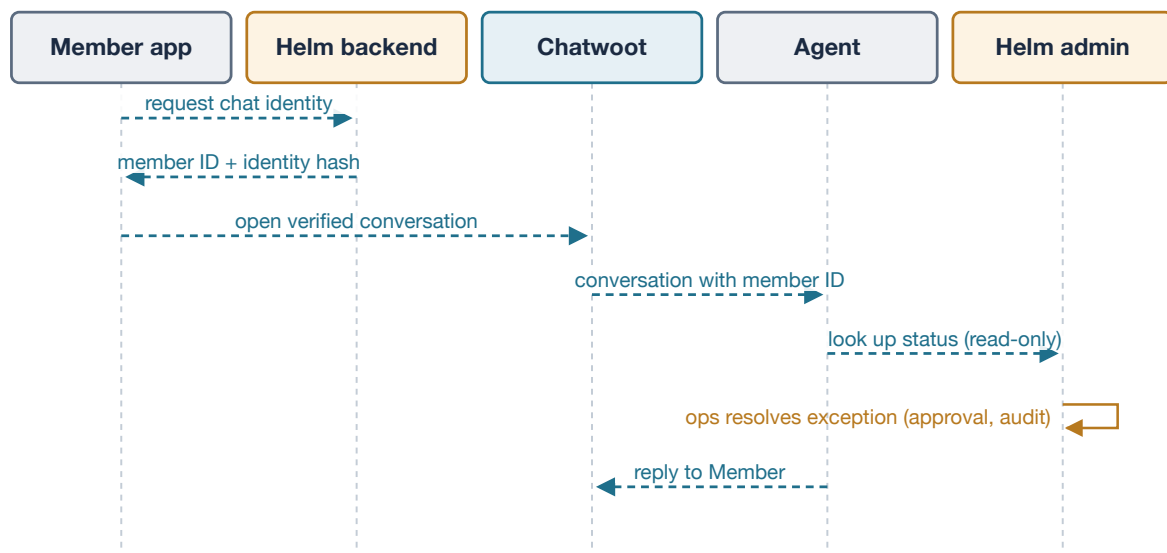


Figure 9: Support and admin exception resolution.

11.1 Flow matrix

Flow	Origin → destination	Minimum data	Authoritative record	Deduplication key	Failure and retry	User sees
Signup and referral	App → Helm → PillarsHub	Privy DID, wallet address, invite code, Sponsor member ID	PillarsHub tree (accepted edge)	Helm member ID (PillarsHub customer ID)	Look up before create; backoff; dead letter to ops; nightly tree reconciliation	“Referral pending” then “confirmed”
Wallet funding	Exchange → chain → Helm indexer	Chain, tx hash, log index, amount, from, to	Chain	(chain, tx hash, log index)	Re-index from last safe block; reorg handling by confirmation depth	Pending → confirmed balance

Flow	Origin → destination	Minimum data	Authoritative record	Deduplication key	Failure and retry	User sees
Vault deposit	App → Onyx contracts; indexer → Helm	Request ID, amount, wallet, terms version	Chain (queue and shares)	(chain, queue address, request ID)	Reverted transactions shown with reason; stale pending alert; Onyx API cross-check	Approval → requested → processing → position
Eligible MLM volume (Phase 1 or if approved)	Helm → PillarsHub sources	Member node, executed amount, date	Helm business event; PillarsHub source	Source externalId = chain:tx:log	Look up by externalId on conflict or timeout	Not shown in Phase 0
Game to Points	Game service → Helm	Event ID, member ID, Points, rule ID	Points service	Event ID	Signed events; replay protection; daily limits	Points balance and history
Redemption	App → Onyx; indexer → Helm	Request ID, shares, destination	Chain	(chain, queue address, request ID)	Awaiting-liquidity alert; cancellation after minimum duration	Requested → awaiting liquidity → paid
Support	App → Chatwoot; agent → Helm admin	Member ID, identity hash, non-sensitive status	Chatwoot (conversation); Helm (actions)	Conversation ID	Widget reset on logout; signed Chatwoot webhooks logged	Chat thread

Unavailable providers: if PillarsHub is down, signups proceed with the referral pending and sync later. If the indexer lags, balances show “updating”. If Onyx’s API is unavailable, Helm relies on chain reads. If the game service is down, play is disabled and no Points are lost.

12 Genealogy and Phase 1 readiness

12.1 Minimum data model

Table	Key fields	Purpose
member	helm_member_id, status, created_at, country attestation	Stable identity
provider_identity	helm_member_id, provider (privy, pillarshub, chatwoot, game), external_id, linked_at	Explicit mappings
wallet	helm_member_id, chain_id, address, source (embedded), first_seen_at, screening_status, allowlist_status	One vault wallet per Member
referral_request	id, helm_member_id, sponsor_member_id, invite_code, channel, captured_at, status, rejection_reason	Pending and rejected referrals
sponsor_edge	id, member_id, sponsor_member_id, effective_at, accepted_at, provenance (signup, import, correction), pillarshub_sync_status, superseded_by	Full graph at every depth
edge_correction	id, edge_id, old_sponsor, new_sponsor, reason, requested_by, approved_by, approved_at	Audited changes
business_event	id, type (funding, deposit_executed, redemption_executed, points_awarded), member_id, amount, asset, chain reference, occurred_at, idempotency_key	Phase 1 volume backfill and reporting
integration_outbox	id, target, payload hash, attempts, status, last_error	Reliable sync

Only data with an identified purpose is collected. No Sponsor or member data is written on-chain.

12.2 Rules

- **Self-referral:** rejected at capture (same member, same wallet, same verified email or phone).
- **Invalid or missing Sponsor:** attach to the company root account with provenance “no sponsor” and flag for review (decision D8).
- **Cycles:** impossible for new Members; checked on every correction and import.
- **Sponsor changes:** never self-service. Admin-only, with reason, second approval and audit entry, then a PillarsHub node update and read-back. Recommended: disable customer movements in the PillarsHub tree configuration so the two systems cannot drift.
- **Account merges:** admin process; keep the earliest accepted edge; re-point child edges with audit entries; retire the duplicate identity.
- **Existing-user import** (only if FirstAlphaWave or TaQUANT members must move): import Sponsors before children, with original signup dates and provenance “import”; there is no bulk API, so import record by record. Verified.

12.3 Proving correctness

- **Full export:** page through every node of the tree with an as-of date, not the downline endpoint, which is capped at 10 levels. Verified.
- **Nightly reconciliation:** compare Helm’s accepted edges with PillarsHub nodes (count, edge-by-edge upline match, checksum). Any mismatch is an exception.
- **Staging test before launch:** a synthetic tree at least 15 levels deep and wide enough to page, created, exported and compared.
- **Phase 1 snapshots:** PillarsHub accepts an as-of date and keeps dated placements, and Helm keeps effective timestamps, so historical genealogy can be reconstructed. No separate snapshot store is needed in Phase 0.

12.4 Volume in Phase 0

PillarsHub calculates Commissions in real time from posted volume. Verified. Posting vault deposits into a commissionable volume type would show pending Commissions to Members. **Recommended:** record executed deposits and Redemptions as business events in Helm only; in Phase 1, backfill PillarsHub with

the original dates once the plan is approved. Whether PillarsHub accepts backdated volume for closed periods is to be confirmed in staging.

13 Backlog, dependencies and schedule

13.1 Prioritised backlog

Priority	Work items	Depends on
Start immediately	App shell and design system; Privy login and embedded wallet; member DB and ledger; referral capture and validation; PillarsHub staging sync with read-back; chain indexer (Sepolia); Onyx SDK deposit and redemption on Sepolia; Chatwoot widget with identity validation; admin skeleton; screening hook with free sanctions API; CI, environments, secrets	Provider test access
Thin slice (weeks 1–3)	End-to-end demo on test networks, including one game-to-Points flow	Game API docs
Before live funds	Vault terms and disclosures with versioned acceptance; allowlist automation; reconciliation jobs and alerts; exception queues; redemption stages and target; country eligibility; Member terms and privacy notice; production environments; security review fixes; operations runbook and rehearsal	Strategy, MLA, ownership, legal answers
Deferrable	Integrated top-up; additional chains and assets; Commission accrual and payouts; Telegram and WhatsApp; Chatwoot audit logs and SSO; commercial transaction monitoring; native mobile	Business demand

13.2 Critical path

1. Meeting decisions D1–D8.
2. **Vault readiness:** strategy and Manager → vault configuration (asset, async queue, external allowlist, non-transferable shares, fees) → MLA signature → production deployment → ownership handover to the Business multisig.
3. **Legal answers:** entity, launch countries, KYC tier, vault terms.
4. **Security review** of Helm's integration and vault configuration.
5. **Operations rehearsal:** valuation, queue execution and Redemption liquidity on a small amount.
6. PillarsHub production launch slot (Tuesday to Thursday).

Engineering work runs in parallel with items 2–3; the launch date is set by whichever of them finishes last.

14 Launch and operational readiness

14.1 Gate for accepting real funds

All must be true; none is waived to meet a date.

1. Vault strategy, Manager, fees and Member terms approved by Business and counsel.
2. MLA signed; deployed version and audit coverage confirmed by Enzyme; upgrade governance documented.
3. Vault Owner multisig in place; Admin and list-owner keys in managed custody; no Owner or Admin key in Helm's backend.
4. Deposits and Redemptions proven on production with a small operator-funded amount.
5. Daily reconciliation passing; monitoring and alerts live; incident contacts at each provider.
6. Screening and eligibility live for the counsel-approved tier.
7. Security review completed and high-severity findings fixed.
8. Written provider acceptance of the business model (Privy, Enzyme, PillarsHub).
9. Support runbook and escalation procedure, including handling of flagged users.

14.2 Security boundaries

- **Audit coverage.** ChainSecurity audited the Onyx core (December 2025), the cross-chain wallet (May 2026) and the Chainlink compliance integration (July 2026). One deployment helper added after the last audit appears in no audit scope found. Verified. Enzyme must confirm which components Helm's vault uses and that they are covered.
- **Upgrade risk.** Enzyme's global owner can upgrade all vault contracts; the audit describes this role as able to "fully drain the system", and no timelock was found. Verified. The MLA must cover notice and governance.
- **Gas sponsorship.** Privy's native sponsorship upgrades Member wallets with EIP-7702; the delegation contract is part of the security review. Verified.
- **Unsigned PillarsHub webhooks.** Treat them as hints and re-fetch through the API. Verified.
- **Test and production separation** for every provider, with separate credentials and no production keys on developer machines.

14.3 Compliance responsibility matrix

Accountable functions are proposals; named owners are TBD.

Control	Applicability question	Accountable (proposed)	Cyclone work	Provider capability relied on	Launch impact
Entity and custody classification	Does any entity have "control" of Member assets (vault roles, allowlist key)?	Business + counsel	Document signing authority and key flows	Privy self-custody (verified); Onyx roles (verified)	Blocker
Launch countries and geographic restriction	Which countries are allowed?	Business + counsel	Country attestation, IP checks, block list	None native	Blocker
Vault product characterisation	Is the vault offering a regulated product in launch countries?	Business + counsel	Terms, fee and risk screens with acceptance record	Onyx strategy and fees (TBD)	Blocker
KYC tier	Is any no-KYC tier allowed; at what thresholds?	Business compliance + counsel	Tier fields and gates now; vendor integration if required	None native (Privy KYC via Bridge unavailable)	Before funds
Sanctions: addresses	Which lists and when?	Business compliance	Screen wallet, funding source, Redemption destination	Chainalysis free API (verified, discovery)	Before funds

Control	Applicability question	Accountable (proposed)	Cyclone work	Provider capability relied on	Launch impact
Sanctions: persons	Is name screening required?	Business compliance + counsel	Hook into identity tier	Vendor TBD	Before funds if required
Vault admission	Only eligible Members may deposit	Business + Enzyme	Allowlist automation after checks	Onyx deposit allowlist (verified)	Before funds
AML monitoring and escalation	What monitoring and reporting duties apply?	Business (compliance officer)	Thresholds, exception queue, case log	None native	Before funds (manual minimum)
Flagged users and transactions	What may Helm lawfully refuse or hold?	Counsel + Business	Restricted states in Helm; no assumed freezing power	Onyx admin discretion and policies (verified)	Before funds
Terms, privacy, consent, retention	Which notices and data-sharing consents?	Business (legal)	Versioned consent; data map per provider	Chatwoot Cloud hosted in the US (verified)	Before funds
Referral model and Points	Are referral rewards and Points disclosed and lawful?	Business + counsel	Rules pages; no earnings claims; Points separate from money	PillarsHub, game service	Before funds (disclosures)
Provider acceptance	Do providers accept the model on full disclosure?	Business	Avoid Stripe and Bridge features in Privy	Written confirmations	Before funds

15 Implementation decisions and feasibility gaps

Decision	Recommended answer	Evidence	Unresolved dependency	Responsible function (proposed)	Launch impact
Network and asset	Arbitrum One, native USDC; Sepolia for tests	Onyx and Privy support Arbitrum; Onyx test deployment is on Sepolia (verified)	Enzyme confirms production chain and asset	Business + Tech	High
Wallet model	User-owned Privy embedded wallet, no app signer, sponsored gas, export enabled	Privy docs (verified)	Privy accepts the business model; volume limit above \$1M per month	Tech	Medium
Vault configuration	Async deposit queue; external allowlist owned by a dedicated list-owner key; non-transferable shares; one asset	Onyx contracts and docs (verified)	Enzyme confirms external list ownership at deployment	Business + Enzyme	High
Vault roles	Business Safe multisig as Owner; separate Admin wallet; limited admin for valuation updates	Onyx roles (verified)	Business signers; Manager appointment	Business	Blocker
Redemption target	Publish an operational target; show stages	No protocol SLA (verified)	Liquidity policy from Manager	Business	High
Genealogy authority	PillarsHub, with Helm pending requests and reconciled copy	Customer and tree APIs (verified)	Auto-placement and correction method in staging	Tech + PillarsHub	Medium
Phase 0 Commissions	None accrued or paid; events kept in Helm for backfill	Real-time calculation and separate payout step (verified)	Terrel's PRD; Business confirmation	Business	Medium
Funding path	Direct crypto; no top-up provider	Paymenture publishes no assets, countries or fees and bars "pyramid schemes" (verified)	Business requirement	Business	Low
Points	No cash value; not linked to deposit amounts	No game docs supplied	Game API documentation; counsel review	Business + Cyclone	Medium
Identity checks	Gates built now; tier set by counsel; address screening from day one	No KYC in stack (verified)	Counsel decision	Business + counsel	High

What would change these defaults: a requirement for integrated top-up at launch adds a payment provider to the critical path; a requirement to pay three-level Commissions at launch adds money-out integration and payout controls; counsel requiring full KYC adds an identity vendor and a gate before vault access.

Appendix A: Evidence

All links were read on 8 October 2026.

PillarsHub: [API authentication](#) · [Access tokens](#) · [Swagger UI](#) · [Customer API spec](#) · [Commission API spec](#) · [Webhooks](#) · [Webhook topics](#) · [Invoice dates and real-time commissions](#) · [Launch timing protocol](#) · [Money-out integrations](#) · [Terms](#)

Privy: [Pricing](#) · [Security FAQ](#) · [Supporting users](#) · [Webhooks](#) · [React quickstart](#) · [Acceptable use policy](#)

Enzyme Onyx: [Product and pricing](#) · [Architecture](#) · [Deposit control and allowlists](#) · [Redemptions](#) · [User roles](#) · [Deployments and upgrades](#) · [Risks and limitations](#) · [SDK](#) · [Public API](#) · [Contracts](#) · [Audits](#)

Chatwoot: [Developer docs](#) · [Pricing](#)

Funding and compliance: [Paymenture](#) · [Bridge developer agreement](#) · [Chainalysis sanctions oracle](#) · [FATF 2021 guidance](#)

Detailed notes with every source and quotation are in `work/notes/01-pillarshub.md`, `02-enzyme-onyx.md`, `03-privy.md` and `04-chatwoot-funding-compliance.md`.

Appendix B: Assumptions and gaps

- **Terrel's PRD was not supplied.** Not validated against it: Commission rules and depth, rank definitions, required back-office features, any requirement for Members to see PillarsHub's back office, any existing-member migration.
- **Game and Points:** no API documentation was supplied. Needed: account linking method, event schema and signing, duplicate and replay protection, abuse controls, Points rules engine, whether Points can be redeemed for anything.
- **Not verified with providers:** PillarsHub auto-placement, depth limits, sponsor correction method, token permission granularity, rate limits, 409 semantics, pricing and security certifications; Privy paid-tier volume limits and business-model acceptance; Enzyme lead times, deployed version, external allowlist ownership, commercial terms; Chatwoot webhook retry behaviour.
- **Pricing:** Enzyme's public price differs from the correspondence; PillarsHub pricing is not public.
- **Team capacity, budget and launch cohort** were not supplied; schedule ranges assume about five engineers.

Appendix C: Questions for providers

PillarsHub

1. Does creating a customer with an enroller place the node in the Unilevel tree automatically, with the upline equal to the enroller?
2. Is genealogy depth unlimited? Does listing tree nodes without node IDs return every node, and what is the page-size maximum?
3. What is the supported, audited way to correct a Sponsor? Can customer movements be disabled?
4. What does a 409 on posting a source mean? Is externalId unique per environment? Are duplicate customer IDs rejected?
5. Can a source group exist that no bonus uses? Is backdated volume accepted for closed periods?
6. Can access tokens be limited by area and deny deletes? What are the rate limits?
7. Can webhooks be signed? Is there a placement or source topic?
8. Please confirm in writing that API-based reconciliation of genealogy is permitted under the Terms, and explain the GLBA disclaimer in this context.
9. Security documentation, data residency, pricing and launch-slot booking.

Privy

1. Do you accept this business model on full disclosure, and do embedded wallets, gas sponsorship and token verification require any Stripe or Bridge account?
2. How is the \$1M monthly transaction volume measured, and what applies above it on paid plans?
3. Can production webhooks or an SLA be added without Enterprise?
4. Security review material for the EIP-7702 delegation used by gas sponsorship.

Enzyme

1. Upgrade governance: who controls upgrades, with what notice, and can Helm's vault opt out?
2. Deployed version and addresses; audit coverage of every component Helm uses, including the deployment helper.
3. Confirm Sepolia for the test vault, Admin App access there, and API indexing of Sepolia.
4. Production chain and recommended deposit asset for Arbitrum.
5. Can an external allowlist owned by a Helm list-owner key be attached at deployment? Indexer delay?
6. Configure shares as non-transferable.
7. Redemption: notice-period or gate modules; recommended cadence; handling a batch that reverts on one request.
8. Valuation automation: lead time, cost, data-provider subscriptions.
9. Commercial terms (deployment fee, AUM option vs public plan, \$6,000 minimum), MLA draft, onboarding form, lead times.
10. Can Enzyme's default investor interface be disabled so Members use only Helm?

Chatwoot

1. Webhook retry and failure behaviour.
2. Data residency options for Cloud; audit logs below Enterprise.

Cyclone game team

1. API documentation, event signing, replay protection, abuse controls and Points rules engine.